Class XII Session 2025-26
Subject - Mathematics
Sample Question Paper - 7

Time Allowed: 3 hours

General Instructions:

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A,.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 Assertion-Reason based questions of 1 mark each.
4. Use of calculators is not allowed.

Section A

1. Let A be a skew-symmetric matrix of order 3. If |A| = x, then $(2023)^x$ is equal to: **[1]**

   a) $(2023)^2$ b) $\frac{1}{2023}$

   c) 2023 d) 1

2. If A is a square matrix of order 3 and |A| = 6, then the value of |adj A| is: **[1]**

   a) 6 b) 216

   c) 27 d) 36

3. The value of $\begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix}$ is **[1]**

   a) 1 b) x + y + z

   c) 0 d) 2(x + y + z)

4. If A is a square matrix such that $A^2 = I$, then A^{-1} is equal to **[1]**

   a) 0 b) A + I

   c) 2A d) A

5. A line making angles $45^0 and 60^0$ with the positive directions of the axis of x and y makes with the positive **[1]**

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direction of Z axis , an angle of

a) 45^0

b) 60^0

c) $60^0 or 120^0$

d) 120^0

6. The Integrating Factor of the differential equation $(1 - y^2)\frac{dx}{dy} + yx = ay\,(-1 < y < 1)$ is **[1]**

a) $\frac{1}{1-y^2}$

b) $\frac{1}{y^2-1}$

c) $\frac{1}{\sqrt{1-y^2}}$

d) $\frac{1}{\sqrt{y^2-1}}$

7. In an LPP, if the objective function z = ax + by has the same maximum value on two corner points of the feasible region, then the number of points at which z_{max} occurs is: **[1]**

a) 0

b) 2

c) infinite

d) finite

8. If i, j, k are unit vectors along with three mutually perpendicular directions, then: **[1]**

a) $i \cdot j = 1$

b) $\hat{i} \cdot k = 0$

c) $i \times k = 0$

d) $i \times j = 1$

9. $\int 2^{x+2}$ dx is equal to: **[1]**

a) $\frac{2^{x+2}}{\log 2} + C$

b) $2^{x+2} + C$

c) $2 \cdot \frac{2^x}{\log 2} + C$

d) $2^{x+2} \log 2 + C$

10. If $\left|A^{-1}\right| = \frac{1}{k|A|}$, where A is a 3×3 matrix, then the value of k is: **[1]**

a) $\frac{1}{8}$

b) 2

c) 8

d) $\frac{1}{2}$

11. The region represented by the inequation system $x, y \geq 0, y \leq 6, x + y \leq 3$ is **[1]**

a) bounded in second quadrant

b) unbounded in first and second quadrants

c) unbounded in first quadrant

d) bounded in first quadrant

12. The position vectors of three consecutive vertices of a parallelogram ABCD are $A(4i + 2j - 6k)$, B $(5i - 3j + k)$ and $C(12i + 4j + 5k)$. The position vector of D is given by **[1]**

a) $-11i - 9j + 2k$

b) $-3i - 5j - 10k$

c) $11i + 9j - 2k$

d) $21i + 3j$

13. If d is the determinant of a square matrix A of order n, then the determinant of its adjoint is **[1]**

a) d^{n+1}

b) d^{n-1}

c) d^n

d) d

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14. The number of arbitrary constants in the general solution of a differential equation of fourth order are: **[1]**

a) 4 b) 3

c) 1 d) 2

15. In $\triangle ABC$, $\overrightarrow{AB} = i + j + 2k$ and $\overrightarrow{AC} = 3i - j + 4k$. If D is mid-point of BC, then vector $\overrightarrow{AD}$ is equal to: **[1]**

a) $2i - 2j + 2k$ b) $i - j + k$

c) $4i + 6k$ d) $2i + 3k$

16. If $x = \log(1 + t^2)$ and $y = t - \tan^{-1} t$, then $\frac{dy}{dx}$ is equal to **[1]**

a) $e^x - y$ b) $e^x - 1$

c) $\frac{\sqrt{e^x} - 1}{2}$ d) $t^2 - 1$

17. The cartesian equation of a line is given by $\frac{2x-1}{\sqrt{3}} = \frac{y+2}{2} = \frac{z-3}{3}$ **[1]**

The direction cosines of the line is

a) $\frac{\sqrt{3}}{\sqrt{55}}, \frac{-4}{\sqrt{55}}, \frac{6}{\sqrt{55}}$ b) $\frac{\sqrt{3}}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{6}{\sqrt{55}}$

c) $\frac{3}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{6}{\sqrt{55}}$ d) $\frac{-3}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{6}{\sqrt{55}}$

18. **Assertion (A):** If manufacturer can sell x items at a price of ₹$(5 - \frac{x}{100})$ each. The cost price of x items is ₹ $(\frac{x}{5} + 500)$. Then, the number of items he should sell to earn maximum profit is 240 items. **[1]**

Reason (R): The profit for selling x items is given by $\frac{24}{5}x - \frac{x^2}{100} - 300$.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

19. **Assertion (A):** A function f: Z → Z defined as $f(x) = x^3$ is injective. **[1]**

Reason (R): A function f: A → B is said to be injective if every element of B has a pre-Image in A.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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